

The Migration Hump Paradox

First Paper

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2026-02-19

1. Sworn Statement

We hereby declare, on our honour, that this report was written entirely by us, without any unauthorised external assistance. It has not been previously submitted for assessment and has never been published, in whole or in part. All passages, phrases, or ideas including tables, figures or charts, that are borrowed from or refer to other sources, are fully and properly cited, without exception.

2. Introduction

Official Development Aid (ODA) is a financial and economic support provided to several developing countries in order to promote economic growth, trade and poverty reduction. However, more informally, ODA also has another goal: to reduce emigration from these countries by giving them more economic resources and raising the living standards which would, in the long term, discourage the population to emigrate. It is precisely this last point that is the subject of much controversy. Indeed, we sometimes see the opposite happening, with an increase in aid preceding an increase in emigration. This phenomenon is known as the *migration hump paradox*. Our project aims to study the impact of Official Development Assistance (ODA) on emigration from developing countries and clarify the existence of this paradox, i.e. to verify, or to refute this theory using statistical and econometric tools.

The population studied consists of 120 low- and middle-income countries according to the World Bank classification. All the estimations of this report are carried out on the single data file *migration_data.gdt*. Our dependent variable is built from the migration rate per 1,000 inhabitants (Y), which is the number of immigrants minus the number of emigrants per 1,000 inhabitants. We will have four explanatory variables:

- X_1 : Net Official Development Assistance in % of GNI (Net ODA)

This is the variable at the heart of our problem. It measures how much a country receives in ODA compared to its gross national income (GNI). Here, in order to capture the long-term effect of Official Development Assistance, we will use data lagged by three years for this variable.

- X_2 : GDP per capita in PPA (GDP PPA)

This variable measures the level of economic development of countries, taking into account the different costs of living, i.e. it measures the real purchasing power. It is an essential variable for analysing migration decisions.

- X_3 : Youth Unemployment (YU)

The youth unemployment rate (18-24) reflects the difficulties young people face in entering the labour market, which can be an important factor in migration decisions – high youth unemployment can increase the incentive to emigrate in search of better economic opportunities abroad.

- X_4 : Political Stability Index (PSI)

This variable is useful and should not be underestimated, as it can explain emigration from countries experiencing instability, conflict or poor governance, where individuals may be motivated to leave their country in search of safety and better living conditions.

gretl : console

```
? open migration_data.gdt
# Explanatory variables
? series X1 = ODA
? series X2 = GDP
? series X3 = Unemp
? series X4 = Stab
# Dependent variable
? series Y1 = mig_rate
? series Y2 = Pop
? series Y3 = (Y1/Y2)*1000
? series Y = sgn(Y3)*ln(abs(Y3))
Série Y1 générée (ID 21)
Série Y2 générée (ID 22)
Série Y3 générée (ID 23)
Série Y générée (ID 24)
```

2.1. Construction of the dependent variable

Before estimating anything, the raw migration ratio has to be transformed. $Y3 = (Y1/Y2) \times 1000$ is extremely dispersed and strongly asymmetric: a handful of very small countries record per-capita balances several orders of magnitude larger than the median, while the bulk of the sample is packed close to zero. Estimating an OLS regression directly on $Y3$ would let these few extreme observations dominate the sum of squared residuals, so the fitted line would describe the outliers rather than the sample.

The usual remedy is a logarithm, which compresses the upper tail and brings the distribution back towards symmetry. A plain $\ln(Y3)$ is not available here, however: the migration balance is negative for a large share of the sample – those are precisely the sending countries at the heart of our question – and the logarithm is undefined for negative values. Dropping them, or taking the log of the absolute value alone, would either amputate half of the sample or erase the distinction between a sending and a receiving country.

The sign-preserving form $Y = \text{sgn}(Y3) \times \ln|Y3|$ solves both problems at once. Taking $|Y3|$ makes the logarithm computable for every country of the sample, and the factor $\text{sgn}(Y3)$ puts the direction of the flow back into the variable, so no information is lost. This is the *log-modulus* transformation discussed by John and Draper (1980), routinely used for variables that are heavy-tailed and take both signs. Its practical benefit is visible in section 3.1: the transformed variable is far better behaved than the raw ratio, and the two migration regimes of the sample become clearly readable on the histogram.

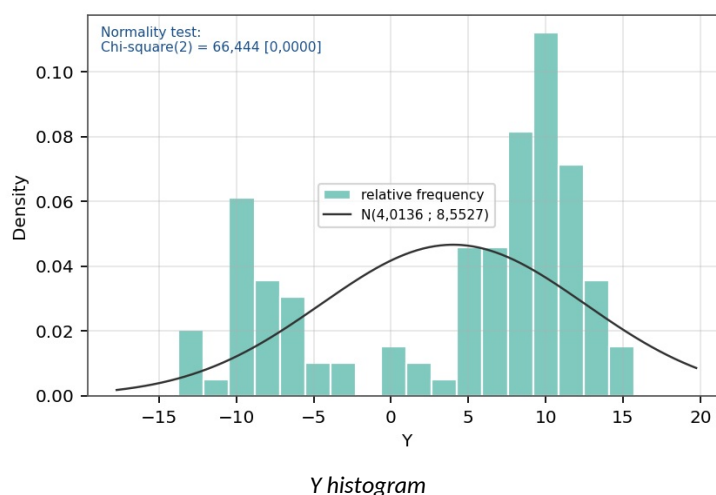
One consequence must be stated explicitly, because it governs how every coefficient of this report is read. Since $|Y3| < 1$ for every country of our sample, $\ln|Y3|$ is always negative, and Y therefore carries the *opposite* sign to the migration balance itself: a net emigration country has $Y > 0$ and a net immigration country has $Y < 0$, with $|Y|$ all the larger as the balance is close to zero. The strict log-modulus variant $\text{sgn}(Y3) \times \ln(1+|Y3|)$ avoids this inversion, but we keep the specification above because it is the one used in both papers, which keeps the two parts of the project perfectly comparable. Y must consequently be read as a transformed indicator of migration intensity, and the

signs of the estimated coefficients are commented throughout with respect to Y , not directly with respect to the number of migrants per 1,000 inhabitants.

The series stored in *migration_data.gdt* are therefore: $Y1 = \text{mig_rate} = (\text{NetMig}/\text{Pop}) \times 1000$, $Y2 = \text{Pop}$, $Y3 = (Y1/Y2) \times 1000$ and $Y = \text{sgn}(Y3) \times \ln|Y3|$. The same definition is used in the Second Paper.

3. Descriptive Statistics

3.1. Endogenous variable



gretl : console

```
? scalar meanY = mean(Y)
Scalaire généré meanY = 4,01362
? scalar sdY = sd(Y)
Scalaire généré sdY = 8,55271
? freq Y --normal
Test de normalité de Y :
  Khi-deux(2) = 66,444 [0,0000]
```

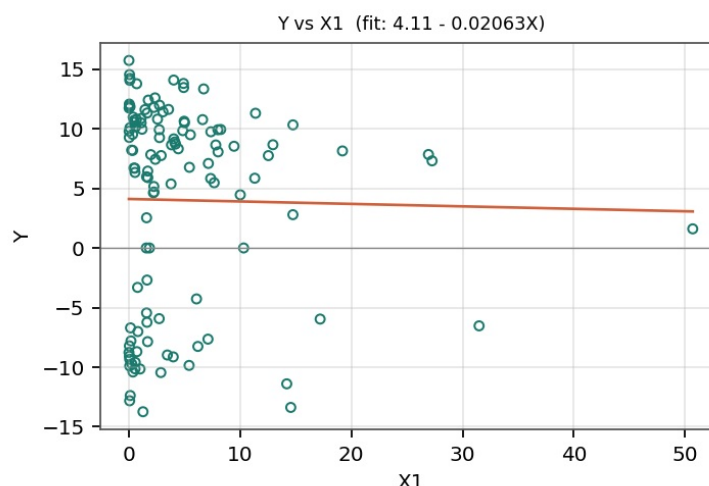
Note: this variable is a logarithmic transformation of the migration ratio, not a growth rate.

The mean of Y is 4.01 and its standard deviation is 8.55. A closer look at the histogram reveals that there are two distinct subgroups, one at around -10 and the other at around 9. This distribution in two distinct groups is confirmed by the fact that the standard deviation is approximately double the mean (indicating that data points are widely dispersed). Given the construction of Y , the positive average means that the sample is dominated by **net emigration** countries: the raw net migration rate is on average -0.09 migrant per 1,000 inhabitants, so the average country of the sample is a sending country, not a host country.

This variable does not follow a normal distribution because it has two peaks (at -10 and 9). The histogram highlights a bimodal distribution confirmed by the normality test (Khi-deux(2) = 66.444, $p < 0.01$). This structure suggests the existence of two distinct migration regimes within the sample, which justifies the subsequent use of a structural break test (Chow test).

3.2. Explanatory variables

a. X1: Net official development assistance



gretl : console

```
? scalar meanX1 = mean(X1)
Scalaire g n r  meanX1 = 4,66088
? scalar sdX1 = sd(X1)
Scalaire g n r  sdX1 = 7,10646
? corr Y X1
corr(Y, X1) = -0,01714366
Sous l'hypoth se nulle d'absence de cor lation :
t(118) = -0,186255, avec une p. critique bilat rale 0,8526
```

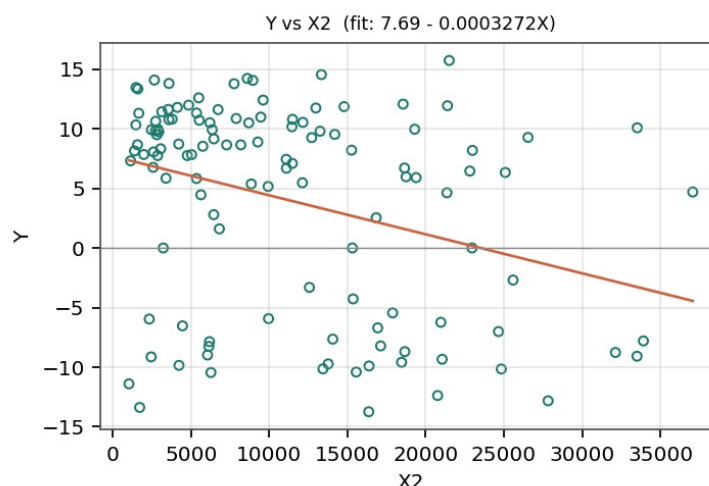
The scatter plot $Y-X_1$ shows the relationship between ODA and the migration indicator. Most countries receive little aid, i.e. the points are concentrated on the left of the graph (when X_1 is on the x-axis).

Given the construction of Y , all the countries above the 0 line ($Y > 0$) are **emigration** countries, and those below the line are immigration countries. It is worth noting that despite massive development aid, emerging countries continue to see part of their population migrate.

On average, countries receive an aid of 4.66% of their GNI. As with migration, the standard deviation is higher than the average. This confirms a certain heterogeneity in the data, with some countries receiving an aid close to 0% of GNI and one outlier receiving nearly 51% of GNI (Tonga).

The descriptive analysis reveals no simple linear correlation between Y and net ODA X_1 , with a coefficient of -0.017 and a non-significant p-value of 0.8526. We cannot therefore reject the null hypothesis of no correlation. Indeed, the bimodal distribution of Y and the dispersion observed on the scatter plot suggest that the effect of aid could be masked by other economic factors or vary across subsamples. Only by switching to multiple regression will it be possible to isolate the specific impact of aid, while the Chow test will verify whether this relationship appears specifically in certain groups of countries, thus validating the complexity of the migration mechanism under study.

b. X_2 : GDP per capita in PPA



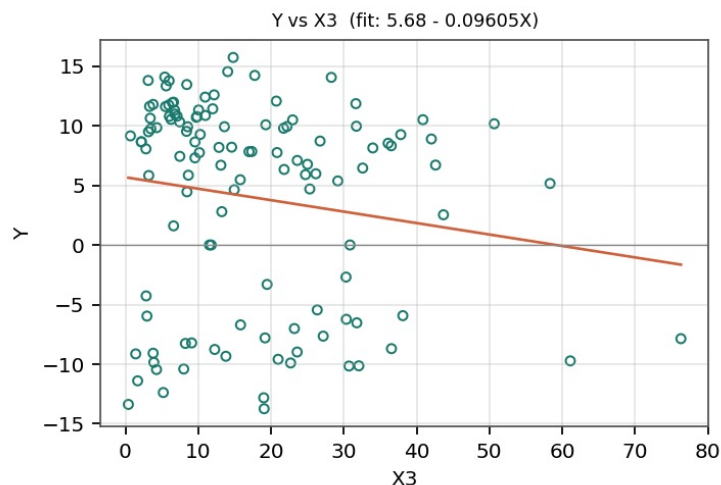
gretl : console

```
? scalar meanX2 = mean(X2)
Scalaire généré meanX2 = 11236,6
? scalar sdX2 = sd(X2)
Scalaire généré sdX2 = 8581,93
? corr Y X2
corr(Y, X2) = -0,32830796
Sous l'hypothèse nulle d'absence de corrélation :
t(118) = -3,77562, avec une p. critique bilatérale 0,0003
```

We note that the average GDP per capita of the countries in the sample is approximately 11,236.6\$. There is a high standard deviation (nearly 76% of the mean). This shows a massive economic divide between the poor and rich countries in our study. The Y-X2 scatter plot shows a negative correlation between the two variables, visible through the downward slope. This negative slope means that as GDP per capita increases, the transformed migration indicator Y decreases: richer countries of the sample are systematically located on the immigration side of the distribution, which confirms the existence of a relationship between a country's wealth and its migration behaviour.

This validates our initial intuition: in richer countries, local economic opportunities (jobs, wages) are better, which reduces the incentive for residents to emigrate in search of a better life elsewhere. The p-value of 0.0003 proves that this trend is statistically very robust. We can also see that the points are very scattered on the left (where the poor countries are located), with extreme situations, some countries experiencing significant mass emigration while others have high immigration. This shows that the migration situation there is unstable and dependent on crises or shocks.

c. X3: Youth Unemployment (18-24)



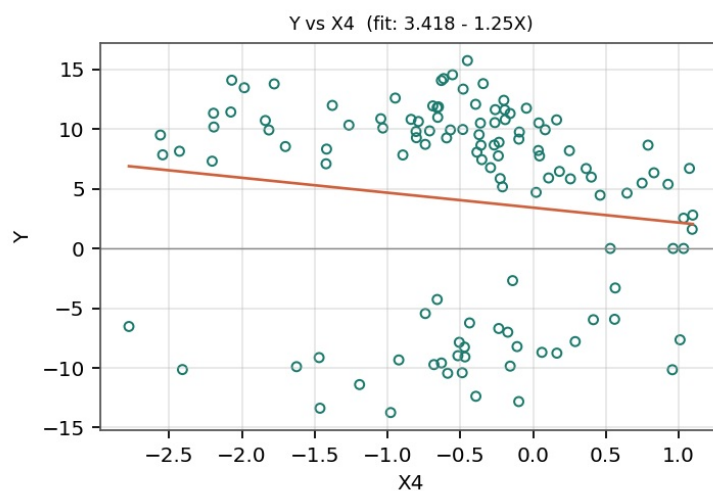
gretl : console

```
? scalar meanX3 = mean(X3)
Scalaire g n r  meanX3 = 17,3523
? scalar sdX3 = sd(X3)
Scalaire g n r  sdX3 = 13,9812
? corr Y X3
corr(Y, X3) = -0,15701321
Sous l'hypoth se nulle d'absence de cor relation :
t(118) = -1,72702, avec une p. critique bilat rale 0,0868
```

The scatter plot shows that there is an inverse relationship, but one that is slightly less clear than for the explanatory variable GDP. The average unemployment rate for the sample is approximately 17.35%, and the standard deviation is approximately 13.98. Once again, we see that the dispersion is high, indicating that our sample is composed of countries with low youth unemployment and countries with employment difficulties for young people.

The fact that this variable is only significant for a 90% confidence interval (p-value of 0.0868) means that although unemployment is a factor encouraging departure, it does not act in isolation and needs to be combined with other variables to reveal its true impact in a multiple regression model.

d. X4: Political Stability Index



gretl : console

```
? scalar meanX4 = mean(X4)
Scalaire généré meanX4 = -0,476905
? scalar sdX4 = sd(X4)
Scalaire généré sdX4 = 0,886323
? corr Y X4
corr(Y, X4) = -0,12950307
Sous l'hypothèse nulle d'absence de corrélation :
t(118) = -1,41871, avec une p. critique bilatérale 0,1586
```

The average is approximately -0.4769, which is negative, meaning that our sample of countries faces greater political or security challenges than normal. This result is logical because our sample includes many developing countries – regions where insecurity is generally highest. We note that the standard deviation is almost twice as large as the absolute value of the mean ($\sigma = 0.886$). This high dispersion of the data suggests that instability is not uniform, with some countries much more stable than others. This heterogeneity is crucial for our model, as it allows us to test whether political insecurity acts as a 'push factor' driving individuals to emigrate.

We note that in the graph showing the Y-X4 scatter plot, the adjustment line is almost flat. However, we can distinguish two subgroups, with one group at the top that is constant and then decreases towards the end, and another subgroup at the bottom that increases gradually. The correlation between X4 and Y is very weak and negative. We cannot directly say that this variable systematically influences the migration indicator in this sample, i.e. we cannot confirm that it has a significant impact on Y. This suggests that political instability may not act alone, but in combination with more powerful economic factors such as GDP.

4. OLS Estimation

4.1. Estimation Results from Gretl

gretl : sortie de code

```
? ols Y const X1 X2 X3 X4
Modèle 1: MCO, utilisant les observations 1-120
Variable dépendante: Y
```

	coefficient	éc. type	t de Student	p. critique	
const	10,5656	1,92172	5,498	2,35e-07	***
X1	-0,259355	0,116378	-2,229	0,0278	**
X2	-0,000395900	0,000103042	-3,842	0,0002	***
X3	-0,0568885	0,0542761	-1,048	0,2968	
X4	-0,194182	0,878703	-0,221	0,8255	
Moyenne var. dép.	4,013619	Éc. type var. dép.	8,552710		
Somme carrés résidus	7387,487	Éc. type régression	8,014925		
R2	0,151323	R2 ajusté	0,121804		
F(4, 115)	5,126269	P. critique (F)	0,000775		
Log de vraisemblance	-417,4757	Critère d'Akaike	844,9514		
Critère de Schwarz	858,8888	Hannan-Quinn	850,6115		
Constante mise à part, la probabilité critique est la plus élevée pour X4					

$$Y = 10.56 - 0.259 X1 - 0.000396 X2 - 0.057 X3 - 0.194 X4$$

After running the OLS estimation of all explanatory variables, we note that only X1 and X2 significantly explain Y. This is consistent with the descriptive analysis, in which each simple regression showed two distinct groups in the scatter plot, indicating that the sample is likely to combine two different

migration regimes.

4.2. Global Significance of the Model

We will check whether this model is useful for explaining the migration indicator. To do so, we will conduct an overall significance test to verify this.

Test hypothesis: $H_0 : SSE = 0$ and $H_1 : SSE > 0$ (unilateral test)

Test Statistics (under H_0): $F = [SSE/p] / [SSR/(n-p-1)]$ which follows Fisher(p, n-p-1) with a confidence level of 95% (with $p=4$ – four explanatory variables)

Rejection Region: $W = (f > F(p; n-p-1), 1-\alpha / H_0)$

gretl : sortie de code

```
? scalar SST = (sd(Y)^2) * (n - 1)
Scalaire généré SST = 8704,71
? scalar SSE = SST - SSR_man
Scalaire généré SSE = 1317,23
? scalar F_final = (SSE / p) / (SSR_man / (n - p - 1))
Scalaire généré F_final = 5,12627
? scalar F_crit = critical(F, p, n - p - 1, 0.05)
Scalaire généré F_crit = 2,45057
--- GLOBAL F TEST ---
SST = 8704,713 | SSR = 7387,487 | SSE = 1317,226
F computed = 5,126269 | F critical = 2,450571
```

We can reject H_0 because $F_{obs} (5.126) > F_{crit} (2.451)$, meaning that the model significantly explains Y. Consequently, the model is globally significant, which validates the continuation of the analysis to identify the individual determinants of migration. We could also have used the critical probability $P(F) = 0.000775 < 5\%$ to demonstrate this.

4.3. Tests on the coefficients

The objective is to test individually whether each explanatory variable X_k has a significant impact on Y. To do this, we perform tests of significance on its coefficients a_k .

Test Hypothesis:

- $H_0 : a_k = 0$ (no effect)
- $H_1 : a_k \neq 0$ (significant effect)

Test Statistics: (under H_0) $T = \hat{a}_k / \sigma_{\hat{a}_k}$ follows $St(n-p-1)$

Rejection region: $W = |t| > T(n-p-1), 97.5\%$

gretl : sortie de code

```
? scalar t_X1 = $coeff[2] / $stderr[2]
? scalar t_X2 = $coeff[3] / $stderr[3]
? scalar t_X3 = $coeff[4] / $stderr[4]
? scalar t_X4 = $coeff[5] / $stderr[5]
? scalar t_crit = critical(t, $df, 0.025)
t-TESTS : X1 = -2,2286 | X2 = -3,8421 | X3 = -1,0481 | X4 = -0,2210 | t* = 1,9808
```

Variable X1 (ODA): The test statistic calculated by Gretl is $t = -2.229$. With 115 degrees of freedom at the 5% threshold, the theoretical critical value (t^*) is 1.981. Since the absolute value of our calculated t (2.229) is greater than 1.981, we reject the null hypothesis. Furthermore, the critical probability

(p-value) is 0.0278, confirming that the variable is significant at the 5% threshold. The coefficient is negative (-0.259), indicating that an increase in development aid is associated with a decrease in Y in this sample.

Variable X2 (GDP): The t-test statistic is -3.842. This value is well above the critical value of 1.981 in absolute terms, leading us to reject the null hypothesis ($a_2 = 0$). With an extremely low p-value of 0.0002, the variable is considered highly significant. The sign of the coefficient is negative (-0.000396), meaning that the higher a country's level of wealth, the lower the transformed migration indicator, which is consistent with classical economic theories on departure factors.

Variable X3 (YU): The test statistic is $t = -1.048$. Comparing the absolute value (1.048) to the critical value (1.981), we see that the result does not fall within the rejection region for the null hypothesis. The associated p-value is 0.2968, which is well above the threshold of 0.05. Therefore, we conclude that the unemployment rate does not have a statistically significant influence on the migration indicator in this specific model. The negative coefficient (-0.057) cannot therefore be interpreted as a reliable structural relationship.

Variable X4 (PSI): Finally, variable X4 has a t-statistic of -0.221. This is the lowest value in the model, indicating a total lack of statistical significance. The p-value of 0.8255 confirms that we cannot reject the null hypothesis ($a_4 = 0$). Despite a negative coefficient of -0.194, we consider that political stability does not explain the variations in migration for the countries in this sample.

In our results, X2 is our most robust variable with a p-value of 0.0002. This indicates that wealth is the central pivot of our model: unlike unemployment (X3) or political stability (X4), which do not appear to be significant in our overall model, GDP is a major determinant.

5. Tests

5.1. Chow Test based on Model 1

The objective of the Chow test is to verify the structural stability of an econometric model. More specifically, in this case, the Chow test is used to test the Migration Hump theory. The sample is split with the dummy variable $D_Rich = (X2 > 11236.6)$, i.e. the threshold given by the average GDP per capita of our descriptive analysis: 68 "poorer" countries ($D_Rich = 0$) and 52 "richer" countries ($D_Rich = 1$).

gretl : sortie de code

```

? series D_Rich = (X2 > 11236.6)
? ols Y const X1 X2 X3 X4
? scalar ssr_p = $ess
? smpl D_Rich == 0 --restrict
Plage d'observations restreinte, 68 observations conservées
? ols Y const X1 X2 X3 X4
? scalar ssr_1 = $ess
? smpl full
? smpl D_Rich == 1 --restrict
Plage d'observations restreinte, 52 observations conservées
? ols Y const X1 X2 X3 X4
? scalar ssr_2 = $ess
? smpl full
? scalar F_chow = ((ssr_p - (ssr_1 + ssr_2)) / k) / ((ssr_1 + ssr_2) / (n - 2*k))
? scalar F_crit = critical(F, k, n - 2*k, 0.05)
--- CHOW TEST ---
SSR group 1 = 3241,358 | SSR group 2 = 3934,270
F chow = 0,6495 | F* (5%) = 2,29687 | p-value = 0,6624

```

Observed value (0.6495) < Critical value (2.2969): we do not reject H_0 , the model is structurally stable. We can therefore conclude that there is no statistically significant difference in the way the variables (ODA, GDP, YU, PSI) influence migration between the poorest and the richest countries of our sample.

The effect of development aid (X1) that we found in the overall model (negative sign) remains the same regardless of the country's level of wealth. Although theory suggests an aid paradox, our Chow stability test confirms the robustness of our initial result: the effect of development aid is uniform, regardless of the wealth level of the country of origin (absence of a statistically detectable 'Migration Hump' in this specification).

5.2. Test for Inclusion of Variables

We will perform an omitted variable test: we will compare a restricted model (we impose the constraint that H_0 is true) to an unrestricted model. The restricted model reduces the freedom of the model by prohibiting it from using ODA to explain migration. Our objective in this test is to see the contribution of ODA (X1) beyond GDP per capita (X2). Hence, our restricted model has a single explanatory variable, GDP (X2), as it represents the variable with the greatest influence on the migration indicator. Our unrestricted model has two explanatory variables, X1 and X2.

The unrestricted model with two variables (Net ODA and GDP per capita):

$$Y = b_0 + b_1 X_1 + b_2 X_2 + e$$

Test Hypothesis:

- $H_0: a_1 = 0$ the restricted model is sufficient to explain the model, i.e. ODA does not provide any significant additional information.
- $H_1: a_1 \neq 0$ aid explains a significant portion of the migration indicator and must be included in the model.

Test Statistics: $F = [(SSR_r - SSR_{ur})/q] / [SSR_{ur}/(n-p-1)]$ follows Fisher $F(q, n-p-1)$

Rejection Region: $W = (F_{obs} > F_{crit} / H_0)$

gretl : sortie de code

```
? ols Y const X1 X2
Modèle 2: MCO, utilisant les observations 1-120 (modèle non restreint)
Variable dépendante: Y
      coefficient    éc. type    t de Student    p. critique
-----
const      9,94684      1,58486      6,276      6,08e-09    ***
X1         -0,253472    0,115736     -2,190      0,0305      **
X2         -0,000422887  9,58382e-05   -4,413      2,29e-05    ***
Somme carrés résidus      7460,616    R2
F(2, 117)      9,755179    P. critique (F)    0,000121

? ols Y const X2
Modèle 3: MCO, utilisant les observations 1-120 (modèle restreint)
Variable dépendante: Y
      coefficient    éc. type    t de Student    p. critique
-----
const      7,69013      1,22338      6,286      5,68e-09    ***
X2         -0,000327190  8,66587e-05   -3,776      0,0003      ***
Somme carrés résidus      7766,466    R2
0,107786

? scalar F_obs = ((ssr_r - ssr_nr) / q) / (ssr_nr / (n - k - 1))
? scalar F_crit = critical(F, q, n - k - 1, 0.05)
? scalar p_val = pvalue(F, q, n - k - 1, F_obs)
--- TEST FOR INCLUSION OF X1 ---
F obs = 4,7964 | F* = 3,9222 | p-value = 0,0305
```

Since the observed statistic (4.7964) is greater than the critical value (3.9222), $F_{\text{obs}} > F_{\text{crit}}$, and the p-value (0.0305) is less than the 5% threshold, we reject the null hypothesis (H_0). We can therefore conclude that adding variable X1 (ODA) significantly improves the quality of the model. The two-variable model is therefore statistically superior to the simple model. Furthermore, the fact that H_0 is rejected proves that Official Development Assistance (X1) provides specific information that GDP (X2) does not contain, because if aid were merely a reflection of a country's poverty, it would add nothing to the model.

In our unrestricted model, the coefficient a_1 of (X1) is -0.2534. For a constant GDP, an increase in development aid is therefore associated with a decrease in Y. Read on the raw migration balance, this means that aid is *not* associated with a reduction of departures in this sample: the countries that receive relatively more aid are those whose migration balance is the most negative. This is precisely the pattern the "migration hump" literature describes, and we will qualify this result in the Second Paper. Hence, we prefer the unrestricted model.

6. Conclusion

gretl : sortie de code

```
? series X1_D_Rich = X1 * D_Rich
? ols Y const X1 X2 D_Rich X1_D_Rich
Modèle 4: MCO, utilisant les observations 1-120
Variable dépendante: Y
      coefficient    éc. type    t de Student    p. critique
-----
const      9,69920      1,60083      6,059      1,79e-08    ***
X1         -0,267967    0,117832     -2,274      0,0248      **
X2         -0,000265438  0,000162018   -1,638      0,1041
D_Rich     -3,37850      3,00694     -1,124      0,2635
X1_D_Rich   0,0193484    0,632203      0,031      0,9756
Somme carrés résidus      7362,433    R2
0,154202
```

In summary, the results demonstrate that the primary drivers of our migration indicator are GDP per capita and Official Development Assistance (ODA), which are the only statistically significant predictors in this sample. The negative coefficient for X_1 (approximately -0.25) indicates that an increase in development aid is associated with a decrease in Y , that is, with a more negative migration balance once the transformation of the dependent variable is taken into account.

Hence, we cannot reject the "Migration Hump" hypothesis, but we cannot formally confirm it either – see the explanations below on the limitations of our model. In fact, by introducing a dummy variable for wealth (D_{Rich}) and an interaction term ($X_1 D_{Rich}$), the analysis tested whether aid affects rich and poor countries differently. The total lack of significance of the interaction term ($p = 0.9756$) proves that the effect of aid is uniform across the sample, regardless of a country's level of development, which is coherent with the Chow test of section 5.1.

We also tested the reliability of the model. Although the adjusted R^2 of the two-variable model (12.83%) may seem modest, it is consistent with the literature on migration determinants in cross-sectional data. Furthermore, the high significance of the F statistic ($p = 0.000121$) confirms that the model is robust overall and that the selected variables are genuine predictors of migration, despite the significant proportion of unobserved qualitative factors.

Finally, we have to address the limits of our model. Two of them must be underlined. First, the dependent variable is a sign-preserving logarithmic transformation of the migration ratio; it magnifies small balances and compresses large ones, so the magnitude of the coefficients cannot be read as an elasticity of the number of migrants. Second, and more importantly, we use net migration instead of emigration numbers. In a context of predominantly South-South migration, aid could fuel the "migration hump" in the poorest countries, causing flows to neighbouring emerging countries. The latter, receiving these flows, show a net migration balance that masks their own emigration. Thus, the difficulty in isolating the paradox in our tests could result from a statistical compensation between the exodus of nationals and regional immigration, rather than a real absence of a migration paradox.

Bibliography

For Y - Net Migration Rate (% population per 1000):

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